TradeLink NG

Digital Trust Infrastructure, AI Discovery & Milestone Escrow for Nigeria's Skilled Informal Economy

Solution Name: TradeLink NG	**Target Geopolitical Zone:** South-West / Nationwide (36 States + FCT)
Sector: Economic Empowerment & Digital Economy	**Current Stage:** Functional MVP / Deployed Production Architecture
Core Innovations: 3-Tier Verification, Escrow, AI Matching, WhatsApp Bot	**Codebase Volume:** 213 Python Modules · 58,950+ Lines of Code

1. Executive Technical Summary & Problem Statement

Over 80% of Nigeria's employment is generated within the informal economy (NBS/ILO data). However, millions of skilled artisans (electricians, plumbers, solar installers, welders, masons) suffer from severe economic invisibility, lack of verifiable qualifications, and persistent payment default risks. Conversely, property owners, corporate site supervisors, and facility managers face unvetted technical quacks, property damage, and abandoned projects.

TradeLink NG solves this systemic trust deficit by providing a battle-tested, end-to-end digital framework combining: (1) **NIN-backed multi-tier credential verification**, (2) **milestone-locked financial escrow**, (3) **AI-powered semantic and visual portfolio matching**, and (4) an **omnichannel WhatsApp Business gateway** designed for low-bandwidth informal workers.

2. System Architecture & Technical Stack

The TradeLink NG platform is engineered with a modular, enterprise-grade Django 5 ASGI architecture running asynchronous task workers, embedded vector embedding models, and multi-provider webhooks:

Architectural Tier	Technology Stack	Functional Responsibilities
Presentation & Omnichannel	• Responsive HTML5 / CSS3 / Vanilla JS • Meta WhatsApp Cloud API • Twilio Programmable Messaging	Dual web and chat interfaces. Allows low-data artisans to receive leads and bid via WhatsApp while employers manage projects on the responsive web portal.
Application & Async Core	• Django 5.x Framework • ASGI Daphne Server • Django Channels (WebSockets)	Handles authentication, role-based access control (Employer, Worker, Admin), live messaging/chats, state transitions for milestones, and escrow management.
AI & Semantic Matching	• FastEmbed (BAAI/bge-base-en-v1.5) • ONNX Runtime (Build-time cached) • OpenAI CLIP ViT-B/32 Embeddings	Computes 768-dim semantic vectors for job descriptions vs. artisan profiles, and visual image vectors for portfolio work verification without expensive GPU overhead.

Data & Storage Tier	• PostgreSQL Database • Redis & Celery Workers • Cloudinary / Media Fallback	Relational integrity for escrow contracts and user records; Redis for asynchronous WhatsApp dispatch and notifications; encrypted media storage.
Financial & Escrow Layer	• Flutterwave API v3 • Webhook Signature Verifier • Virtual Bank Accounts	Processes milestone escrow deposits, handles virtual accounts for seamless transfers, and executes programmatic payouts upon milestone inspection.

3. Core Technical Modules & Workflows

Pillar A: Multi-Tier Verification System (Trust Badges)

To dismantle identity fraud and incompetent workmanship, TradeLink NG operates a 3-tier progressive verification system, granting distinct cryptographic and visual badges on artisan profiles:

Tier Level	Requirements & Verification Source	Rights & Project Scope
BLUE BADGE **Level 1: Identity**	• Verified National Identification Number (NIN) • Phone number OTP confirmation • Live selfie face match	Eligible for residential maintenance tasks, neighborhood repairs, and standard marketplace listings.
GOLD BADGE **Level 2: Certified Trade**	• Blue Badge requirements + • Trade certification (NABTEB, City & Guilds) • Apprenticeship documentation & vetting	Qualified for commercial projects, specialized installations (solar, HVAC, industrial wiring), and preferred recommendation ranking.
GREEN BADGE **Level 3: Corporate Entity**	• Gold Badge requirements + • Corporate Affairs Commission (CAC) certificate • Tax Identification Number (TIN)	Authorized to bid on large-scale enterprise construction tenders, government subcontracts, and multi-million naira projects.

Pillar B: Milestone Escrow & Dispute Resolution State Machine

Our milestone payment engine eliminates 'payment anxiety' for artisans and 'abandonment anxiety' for clients. Work is governed by a strict state-machine flow:

Step 1: Job Milestone Creation	Employer posts project and subdivides compensation into inspectable milestones (e.g. 30% Foundation, 40% Roofing, 30% Finishing).
Step 2: Upfront Escrow Deposit	Employer deposits milestone funds into TradeLink's secure escrow vault via bank transfer or card. Funds are locked.
Step 3: Work Execution & Alert	Artisan receives instant automated notification that funds are secured and commences work with guaranteed payout backing.
Step 4: On-Site Inspection & Release	Client inspects completed milestone on-site. Upon client approval, funds are disbursed directly to artisan's Nigerian bank account within seconds.
Step 5: Dispute Mediation Vault	If specifications are breached, either party can trigger mediation. Milestone remains frozen while human trade arbiters review evidence and resolve payouts.

Pillar C: Omnichannel Conversational Access (WhatsApp Business Gateway)

To ensure complete accessibility for artisans without high-end smartphones or constant web data, TradeLink NG integrates directly with the **Meta WhatsApp Business Cloud API** and **Twilio Webhooks** (implemented in bot/views.py and bot/handlers.py):

- **Instant Job Broadcasts:** Verified technicians receive localized job alerts matching their trade and LGA directly in WhatsApp.
- **Conversational Bidding:** Artisans can reply with their quote and proposed timeframe without leaving the chat app.
- **SMS Fallback:** In rural or low-bandwidth zones, SMS gateways ensure project status and escrow release alerts are delivered reliably.

4. Technical Evidence & Codebase Verification

TradeLink NG is not an unproven concept. The solution is backed by an extensive, battle-tested codebase comprising 10 custom Django applications and production infrastructure:

Django Application	Lines of Code	Key Implemented Features & Technical Capabilities
marketplace	~18,500 lines	Multi-vendor trade tools marketplace, category taxonomy, Jumia/Etsy Apify catalog scrapers, real-time product search, order management, and dispute workflows.
jobs & hiring	~14,200 lines	Dynamic job milestone architecture, worker profile generator, portfolio showcase with CLIP vector embeddings, bid evaluation engine, contract generator.
bot	~6,500 lines	WhatsApp Cloud API webhook handler, Twilio router, stateful session dispatcher (WhatsAppSession), conversational bidding, lead alerts.
verification	~4,200 lines	3-tier verification pipeline, NIN cryptographic matching, document uploads, vetting console, badge rendering.
chats	~3,800 lines	Real-time messaging via WebSockets (Django Channels / ASGI Daphne), attachment handling, instant notifications.
core & users	~11,750 lines	Custom User model, RBAC middleware, global rate-limiting, error logging, FastEmbed model caching, landing and legal pages.
TOTALS	**58,950+ lines**	**213 Python source modules, 10 custom apps, fully containerized on Railway Nixpacks.**

5. Security, Data Privacy & NDPA Compliance Architecture

TradeLink NG enforces enterprise-grade security protocols compliant with the **Nigeria Data Protection Act (NDPA)** and international privacy standards:

Encryption Everywhere	Enforced TLS 1.3 across all HTTP endpoints and webhook consumers. Relational data and documents encrypted at rest using AES-256; passwords hashed via salted PBKDF2/Argon2.
Zero Card Storage	No debit card numbers, CVVs, or bank account PINs touch TradeLink servers. All payment flows are tokenized via licensed, PCI-DSS Level 1 compliant gateway partners (Flutterwave).
Role-Based Access (RBAC)	Strict data segregation. Public clients only see validated badges and aggregate reviews. Private identification (NIN, CAC documents) is strictly restricted to vetted compliance staff.
Testing Isolation	Development and staging environments are logically and physically isolated from production. All testing and model validation utilizes anonymized synthetic datasets.

6. Project Implementation Roadmap & Grant Milestones

Grant funding will be deployed across four milestone-driven phases over the next 12 months to transition from operational MVP to nationwide adoption:

Phase & Timeline	Key Technical & Operational Deliverables	Expected Impact Target
Phase 1 **Months 1–2**	• Integrate direct NIMC-accredited API for instant NIN validation. • Finalize automated virtual account bank verification. • Launch streamlined artisan onboarding UX.	Verification latency cut from 24 hours to <2 minutes.
Phase 2 **Months 3–4**	• Launch localized union drives across Lagos (Alaba, Somolu, Coker) and Abuja. • Onboard and badge initial pilot cohort. • Execute targeted homeowner marketing campaigns.	1,000 verified artisans onboarded; 500 completed escrow jobs.
Phase 3 **Months 5–7**	• Release lightweight Android PWA with offline caching. • Deploy low-data image compression for portfolio uploads. • Expand WhatsApp bot NLP commands in Pidgin & Yoruba.	5,000 active trade workers; 70% retention rate.
Phase 4 **Months 8–10**	• Partner with licensed MFIs to pilot equipment lease-to-own financing. • Embed project-liability and accident micro-insurance. • Launch Trade Credit alternative scoring API.	■250M+ total gross milestone volume disbursed to artisans.
Phase 5 **Months 11–12**	• Nationwide expansion into Rivers, Kano, Oyo, and Kaduna. • Formal partnership with NABTEB and ITF for certified apprentice placement.	15,000+ verified artisans; 30,000+ jobs fulfilled across Nigeria.

7. Project Lead & Verification Contact

Platform Name: TradeLink NG	**Official Website:** tradelink.ng
Primary Email: contact@tradelink.ng	**Technical Stack:** Django 5 · PostgreSQL · Daphne · FastEmbed · Flutterwave